IMPERIAL RESEARCH ARM · KAGGLE QUALIFIER

Name That LLM

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THE TASK

Which of 18 language models wrote this passage?

18

models from six labs: OpenAI, Google, Anthropic, Qwen, Z.ai, Moonshot

2,699

training passages, one per model for each of 150 prompts

~4,000

hidden test passages on prompts never seen in training

BA

balanced accuracy, so every model counts the same

Labs are easy to tell apart. Siblings are not: glm-5.3 against glm-5.3-flash, and gemini-3.7-flash against gemini-3.8-flash. Every model we tried stalls on those two pairs.

STARTING POINT

Validation first, then build on what others had already published.

01

Freeze the folds

Five folds grouped by prompt, seed 42, saved to a file. No prompt sits in both training and validation, as in the hidden test.

02

A plain baseline

A four-view n-gram SVC on host data only: **0.348** cross-validated. Every later change is one row in an experiment log.

03

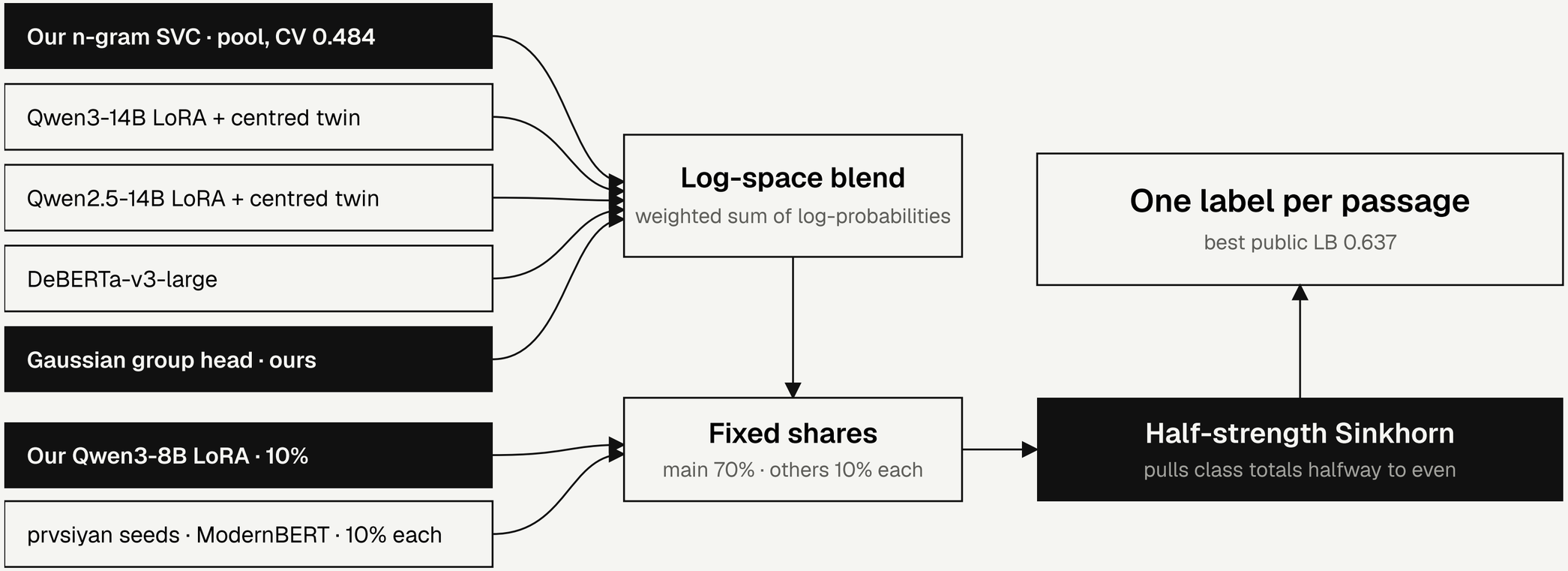
Read, don't fork

The best public blend scored 0.620. We wrote our own loader for its published LoRA weights and added our own members on top.

Rules we kept: train only on Kaggle, everything free and openly licensed, cite every source, no hand-labelling of test data.

APPROACH

Many views, one blend, then balancing.



Filled boxes are ours. Outlined boxes are published weights run through our own loader, with the end token and label order checked.

HYPOTHESES

Our hypotheses, and what happened.

HYPOTHESIS	WHY	RESULT
Style lives in small habits	Function words, punctuation and word shapes change little with the topic.	Four n-gram views: CV 0.348 on host data alone.
Data beats model size	2,699 passages is tiny. Public in-style passages show more of each model's habits.	84k public passages took the SVC to **0.484**.
The prompt hides the author	Test passages come in groups that share a prompt. Take out what a group shares and style is left.	Gaussian group head: **+0.058** on its own view.
The test mix is uneven	Full balancing bets on an even test. A half step hedges.	Half Sinkhorn: **+0.003** on the leaderboard.

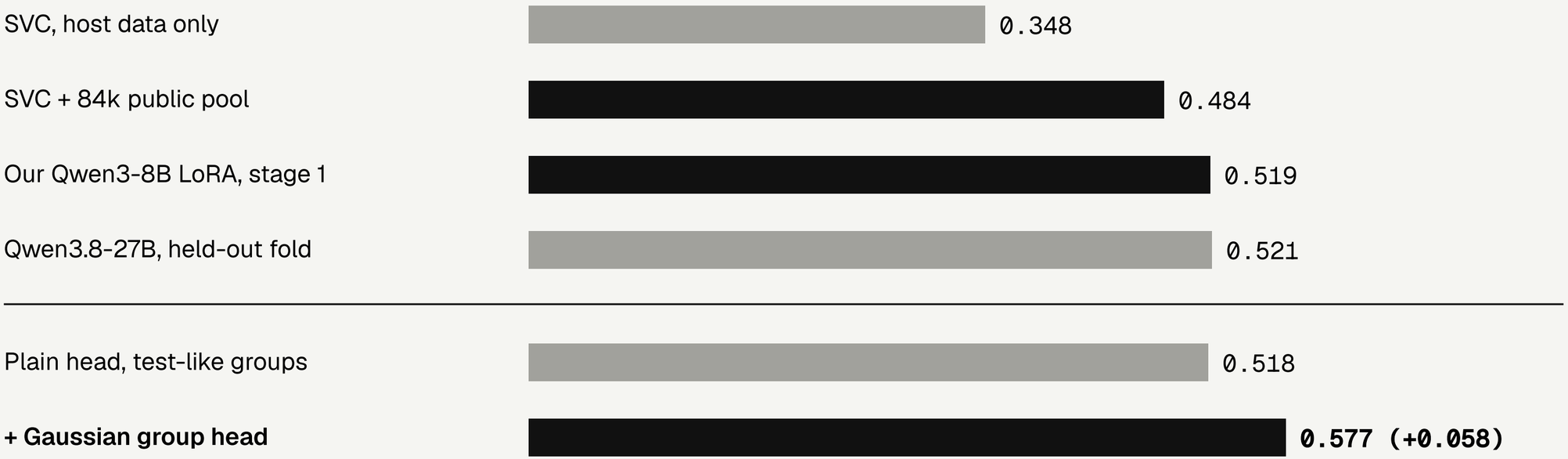
ABLATION · PUBLIC LEADERBOARD

Each probe changed one thing.

PROBE	CHANGE	PUBLIC LB	Δ
1	Our SVC + Nishant Dahal's published spine, full Sinkhorn	0.626	—
2	Sinkhorn at half strength	0.629	**+0.003**
3	+ prvsiyan's two Qwen3-8B seeds at 10%, SVC pool weight 0.25	0.631	+0.002
4	+ Zbigniew Kirchner's in-style Qwen3-8B at 10%	0.627	−0.004
5	Probe 3 + UltraBERT 10% + ATLAS on unsure rows	0.631	±0
6	Peer correction in place of Sinkhorn	0.622	−0.009
7	Probe 3 + our own Qwen3-8B LoRA at 10%	0.631	±0
8	+ Gaussian group head at weight 0.4	0.631	±0
9	Gaussian head at weight 1.0 + ModernBERT at 10%	**0.637**	**+0.006**

ABLATION · HONEST CROSS-VALIDATION

Scores on folds each model never saw.



Flat, within noise: a prompt-centred SVC (+0.003 ± 0.013), pair specialists for the hard pairs (+0.0026), a 122k public pool for the SVC (−0.004). The 27B needs about 3 s a passage, too slow for a 9-hour run.

WHAT DIDN'T WORK

Peer correction: 0.631 → 0.622, our worst probe.

The idea

A public trick: inside each topic cluster, remove the lean the SVC gives every passage, and drop class quotas.

Why we tried it

Its author saw +0.010, on a weaker blend. On paper it targets the same problem as our hypothesis about prompts.

Why it failed

With no quotas, a strong blend drifts towards the models it already favours. On the full public stack it also lost (−0.001).

Lesson: a trick that rescues a weak blend can hurt a strong one. Measure it on your own base.

WHAT THE LEADERBOARD TOLD US

Four ties at 0.631, then a jump to 0.637.

The public slice is small

Probes 3, 5, 7 and 8 each changed real members, and each scored 0.631. Gains of a few thousandths round away, so we steered by cross-validation, not the board.

The Gaussian head needed full weight

At 0.4 it moved too few rows. At 1.0, the weight our cross-validation picked, plus ModernBERT, the score rose to 0.637.

0.637

Best public score (probe 9), up from 0.626 at our first probe and 0.620 for the published spine alone.

WHAT I'D TRY NEXT

More data where we are weakest, and honest numbers for every member.

01

Bigger pool

Retrain on the open 122k pool. Stage 1 alone is about 10 h on two T4s.

02

Five folds each

Fold models for every member, so the blend is fitted on honest scores, not probes.

03

Joint decoding

One Gaussian group model across all members, with one shared prompt effect.

04

The hard pairs

Targeted data for the GLM and Gemini siblings, once the rules on generated data are clear.

Code: github.com/Nabzx/name-that-llm · Every AI session exported in ai-sessions/ · Thanks for watching.